

Uncovering Trends in Police Shootings: A Data-Driven Dashboard

1. Scenario

a. What is the problem you are trying to solve?

In recent years, there has been heightened public interest in understanding the nature, frequency, and contextual circumstances of police-involved shootings in the United States. The core problem is the lack of accessible, consolidated data storytelling that allows the public to visualize patterns across time, geography, and demographics. Our project seeks to bridge this gap by creating a descriptive dashboard that illuminates the landscape of these incidents. It moves beyond simple counts to reveal trends in fatality, armed status, and racial disparities, helping users answer foundational questions of what happened, where, and to whom.

b. Who is the intended audience?

The dashboard is designed for analysts, researchers, policy-makers, and the general public. It supports those seeking to develop an evidence-based understanding of incident patterns to inform further decision-making around policy, training, oversight, and community engagement.

c. Where did your data come from?

The data is primarily sourced from the US Police Shootings dataset on Kaggle. It encompasses critical variables such as date, location (city, state), victim demographics, armed status, threat level, and body camera presence.

d. Who acted as your client and what perspective did they offer?

Our client perspective is that of a Public Policy Research Institute. They offered a perspective focused on descriptive analytics rather than predictive modeling. They required a dashboard that establishes a historical "Source of Truth" to illuminate the landscape of police-involved shootings through clear, accessible storytelling, serving as a springboard for further investigation.

2. Describe the steps taken in the project

a. Analysis

Geographic:

For the geographic mapping and regional analysis component, we focused on translating location-based police shooting data into a set of interactive spatial visualizations using Tableau. After loading and cleaning the dataset, we ensured that geographic fields such as state and city were properly standardized and recognized as geographic dimensions. We then constructed multiple location-focused views, including a filled map showing total shootings by state, a city-level circle map highlighting local clusters, a bar chart ranking the top ten states by shooting rate, and a state-by-year heatmap to examine temporal variation across regions.

The geographic analysis focuses on identifying spatial concentration, regional disparities, and temporal patterns in police shooting incidents across the United States between 2015 and 2020. The filled state map provides a high-level overview of how incidents are distributed nationally, while the city-level circle map offers a more granular view of urban concentration and clustering. The bar chart ranking the top ten states enables direct comparison between high-incident states, and the state-year heatmap reveals how incident intensity changes over time within each state.

To support interactive exploration, filters for race, armed status, and date were added and applied consistently across all geographic views. These filters allow users to dynamically adjust the scope of analysis while maintaining comparability across visualizations.

Demographic:

For the demographic analysis, we first created grouped variables—such as consolidating weapon types into an Armed vs Unarmed category—to enable clearer and more meaningful comparisons across population groups. We then incorporated parameter controls, geographic maps, box plots, and distribution charts to examine temporal trends, demographic patterns, and spatial disparities within the data. Finally, we combined these visual components into a cohesive, interactive dashboard that allows users to filter, compare, and drill down into the key dimensions of police-involved shootings.

We used bar charts, histograms, and box plots to compare distributions and identify which groups experience higher incident counts or distinct age patterns. A geographic map added a spatial perspective, allowing us to observe regional concentrations and state-level disparities.

Temporal:

For the temporal analysis, we focused on identifying month-to-month patterns, long-term trends, and year-over-year fluctuations in police shooting incidents using Tableau. After preparing the dataset and verifying that the date field was properly recognized as a continuous temporal dimension, we constructed a

series of visualizations, including a multi-year monthly line chart to observe recurring patterns, a bar chart breaking down incident counts by race across years, and a year-by-month heatmap to highlight seasonal intensity and temporal variance. The line chart enabled direct comparison of monthly trends from 2015 through 2020, while the bar chart illuminated demographic stability across time periods. The heatmap proved particularly useful for revealing subtle differences in incident frequency from month to month and for contrasting temporal patterns across different years. Together, these visualizations allowed us to explore temporal behavior at multiple scales, from short-term monthly fluctuations to broader multi-year dynamics.

b. Discoveries

Temporal:

Several notable insights emerged from the temporal visualizations. The multi-year line chart revealed that although shooting counts fluctuate from month to month, there is no strong evidence of a consistent seasonal pattern across all years; instead, the data shows moderate, irregular variation that differs from year to year. The year-by-month heatmap reinforced this finding by illustrating distinct pockets of higher activity that occur in different months depending on the year, suggesting that external factors—rather than regular seasonal cycles—may be influencing incident frequency. Meanwhile, the race-by-year bar chart demonstrated that the demographic composition of victims remains relatively stable over time, with White individuals consistently representing the largest group, followed by Black and Hispanic individuals in similar proportions each year. Overall, the temporal analysis highlights persistent demographic patterns, moderate month-to-month variability, and the absence of predictable seasonal cycles.

Geographic:

Several notable patterns emerge from the geographic visualizations. At the state level, the filled map shows that shootings are not evenly distributed across the country, with higher concentrations in large and populous states such as California, Texas, and Florida. This pattern is reinforced by the top-10 bar chart, where these states consistently rank highest in shooting counts during the selected time period.

At the city level, the circle map reveals strong clustering in major metropolitan areas, such as Los Angeles, Chicago, and Oklahoma City, indicating that incidents are highly concentrated in urban environments rather than evenly spread across states. Additionally, the state-year heatmap shows that while high-incident states tend to remain consistently high over time, there are year-to-year fluctuations within individual states, suggesting that trends are not

static. Overall, the analysis highlights persistent regional disparities as well as temporal variation, underscoring the importance of examining both geography and time when interpreting police shooting data.

Demographic:

The data revealed strong demographic disparities, with White and Black victims representing the largest proportions of incidents. The age analysis showed that both armed and unarmed victims tend to fall within a 25–40 age range, though unarmed victims appeared slightly younger on average. Geographically, shootings were heavily concentrated in southern and western states, highlighting clear regional patterns in incident frequency.

c. Challenges encountered and resolutions

One major challenge was managing visual clutter in the city-level cluster map, as large numbers of overlapping points made some regions difficult to interpret. This issue was addressed by adjusting mark size, transparency, and zoom levels, allowing clusters to remain visible without overwhelming the map. Another challenge involved ensuring consistent filtering across all worksheets, which required careful configuration so that race, armed status, and date filters applied uniformly across the dashboard.

Another difficulty arose when calculating demographic percentages over time because Tableau treated Boolean expressions as non-aggregatable dimensions, preventing the use of average calculations. We addressed this by converting the Boolean expression into a numeric 0/1 variable, enabling percentage computation through the `AVG()` function. Finally, the multi-line temporal chart initially appeared cluttered when all years were displayed simultaneously; this was mitigated by refining line colors, adjusting opacity, and improving legend clarity so that year-to-year comparisons were more interpretable.

d. Adjustments Made from the Original Plan

Originally, the geographic analysis was planned to focus primarily on state-level comparisons. However, during development, we found that state-level aggregation alone obscured important local patterns. As a result, we expanded

the scope to include city-level clustering and a state-year heatmap, which provided additional insight into urban concentration and temporal dynamics. This adjustment improved the analytical depth of the dashboard while remaining aligned with the project's descriptive focus.

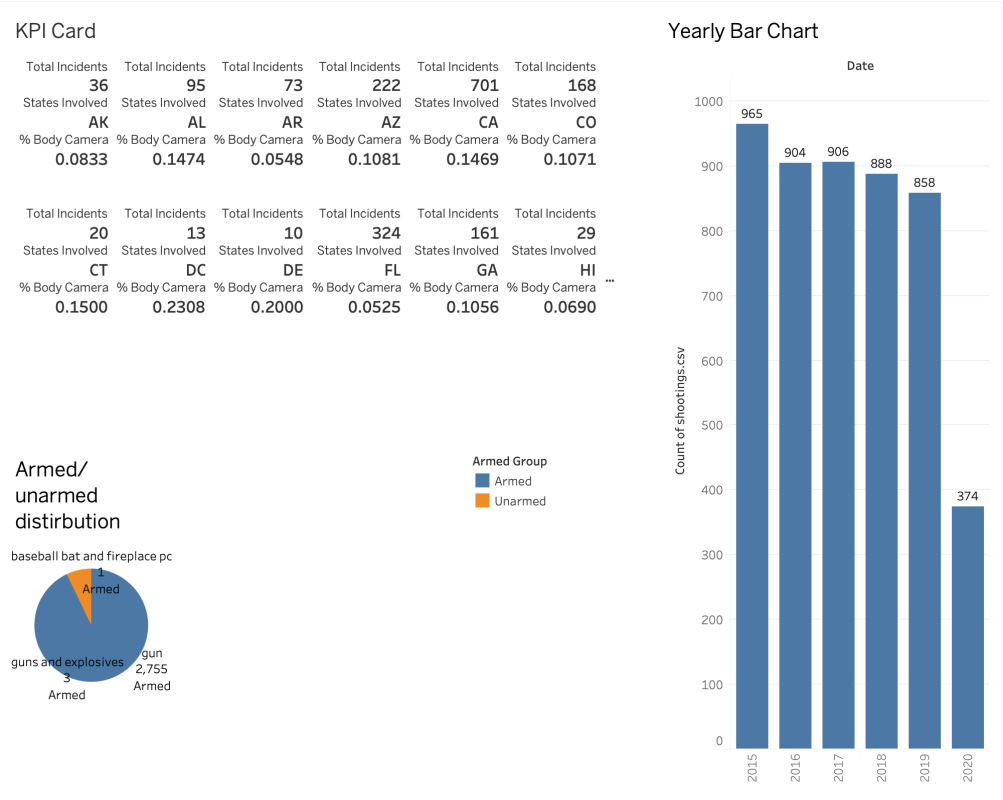
The original temporal analysis plan included building a standard set of line and bar charts to observe temporal behavior, but during development it became clear that these alone did not adequately highlight month-by-month differences across years. As a result, we added a year-by-month heatmap, which offered a more intuitive and visually compact way to compare temporal intensity across multiple years. Additionally, demographic percentage trends were initially planned as line charts but required adjustment due to the need for custom numeric calculations and formatting to properly represent percentages. These modifications improved analytical clarity while staying aligned with the overarching goal of uncovering meaningful temporal patterns.

e. Feedback from Our Client and How We Incorporated Them into Our Final Product

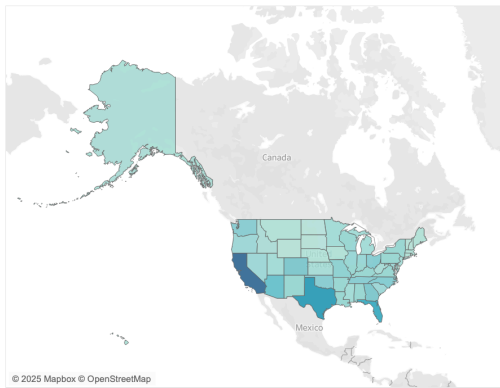
Feedback from the client emphasized the importance of clarity, interpretability, and interactive exploration. In response, we simplified color schemes, reduced unnecessary visual complexity, and ensured that all geographic views shared the same filters. The client also suggested including a temporal comparison component, which led to the addition of the state-year heatmap. These changes improved the dashboard's usability and strengthened its ability to support evidence-based exploration by a broad audience.

Final Dashboards

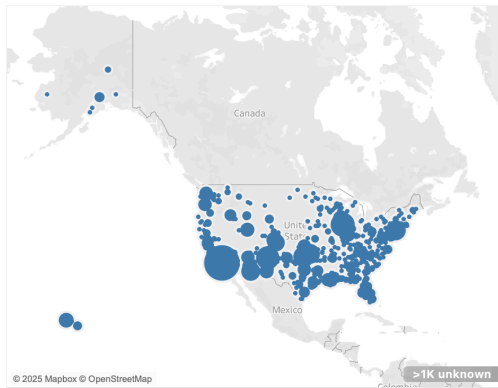
Dashboard Overview: This dashboard presents a comprehensive analysis of 4,895 police-involved shooting incidents across 51 U.S. jurisdictions (50 states + DC). It breaks down key metrics including temporal trends, victim demographics, and armed status. By visualizing this data, we aim to provide transparency and actionable insights into the scale and nature of these incidents, supporting evidence-based policy discussions.



Shooting by State



City-Level Clusters



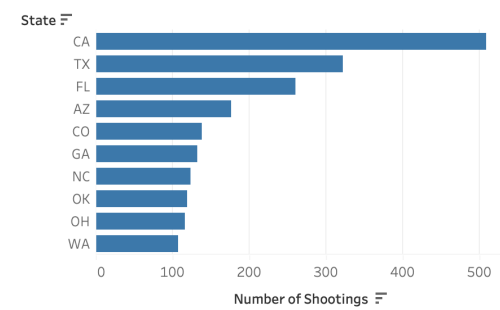
Date
2016/ 2020/
0 1000

Race
(All) ▾

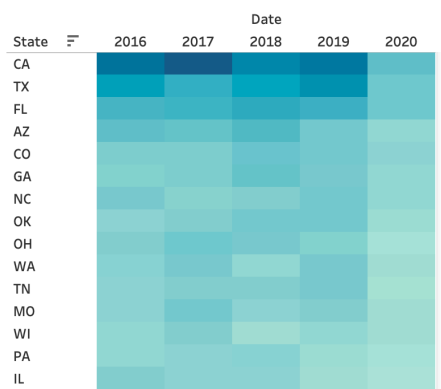
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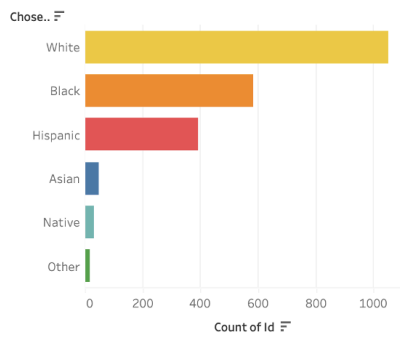
Top 10 States by Shootings



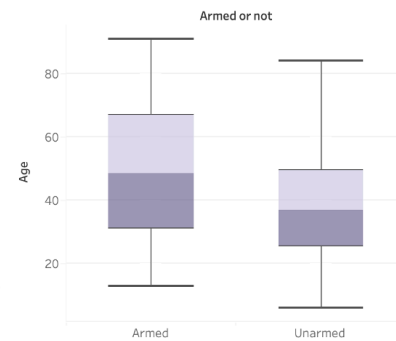
State-Year Heatmap



Shootings by Race or Gender by Year



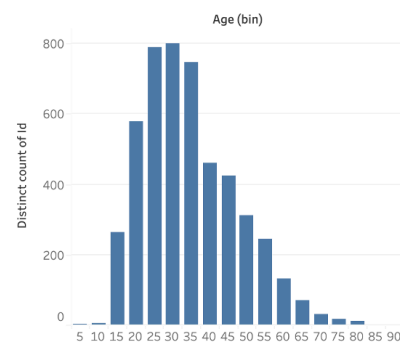
Age Distribution by Armed Status



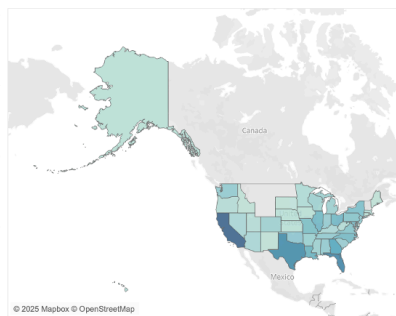
Year of Date
☐ (All)
☐ 2015
☐ 2016
☐ 2017
☒ 2018
☒ 2019
☒ 2020

Chosen Category
☒ Asian
☒ Black
☒ Hispanic
☒ Native
☒ Other
☒ White

Shootings by Age



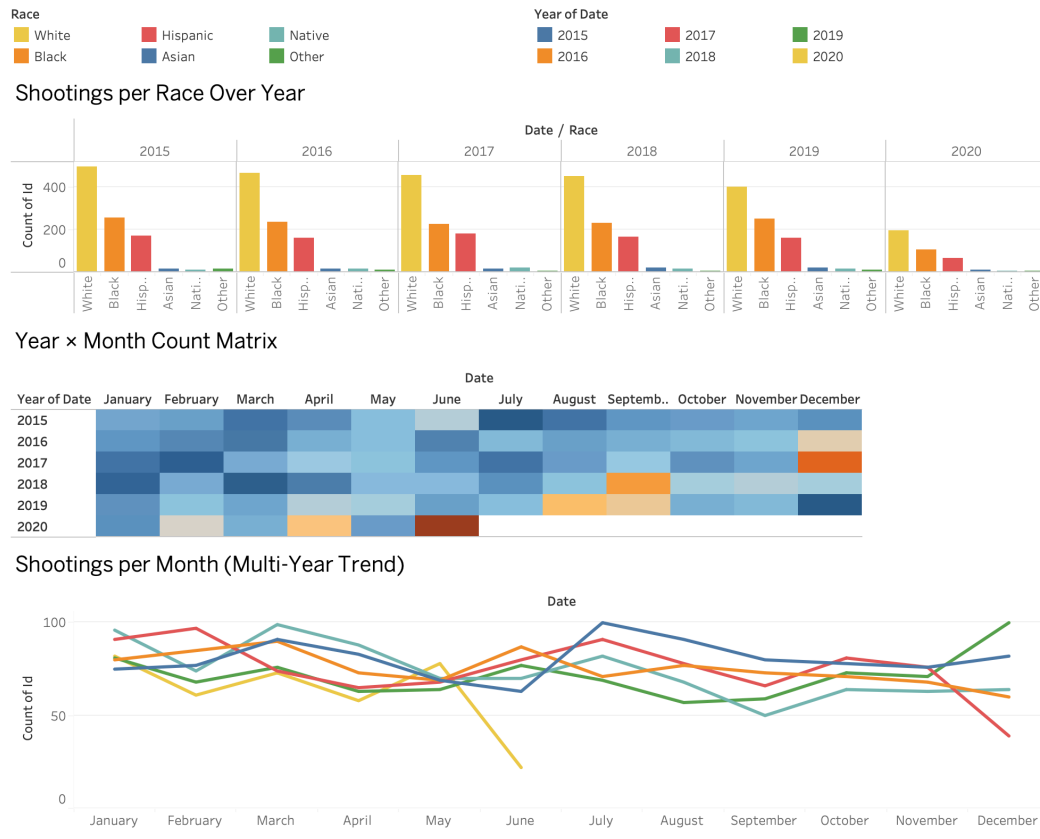
Geographic Distribution of Shootings by Race



Choose Category
Race ▾

Race
☐ (All)
☒ Asian
☒ Black
☐ Hispanic
☐ Native
☐ Other
☐ White

Count of Id
1 151



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